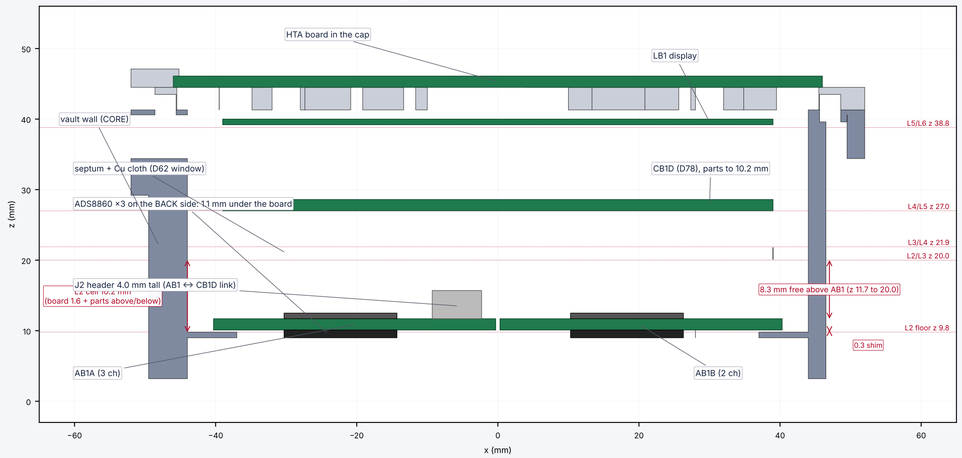
PHI-FOLD 233 receive stage cut-away: why AB1A / AB1B are hard to place and route (MODELED, tool-measured, nothing built)

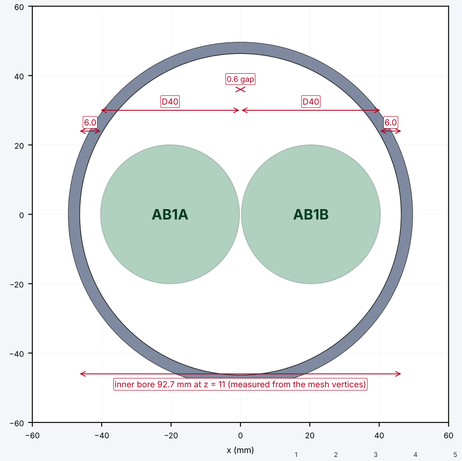
1 Vault half-section (y = 0), cut faces from the current meshes

heel (178 deg) to the left; boards drawn as slabs at the DESIGN_STATE section 3 z-ranges; D40 AB1A/AB1B at x = -/+ 20.3



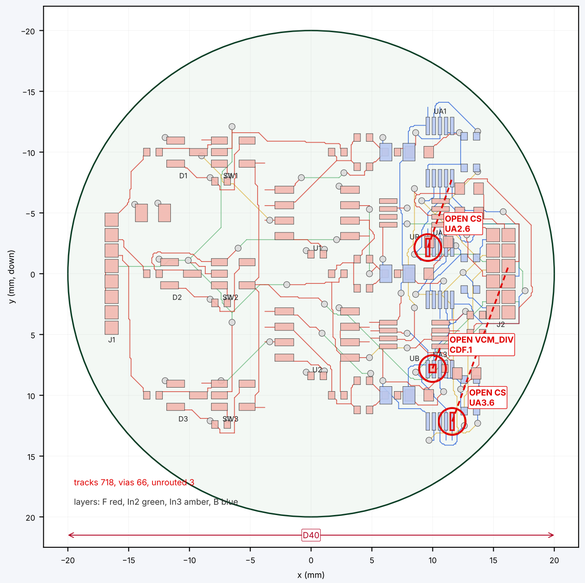
2 Plan section at z = 11 (AB1 level): what the two D40 discs must fit inside

ring = the core wall cut at z = 11; discs AB1A / AB1B at x = -/+ 20.3; the shared strip between them carries J2 and the BLANK / SPI links



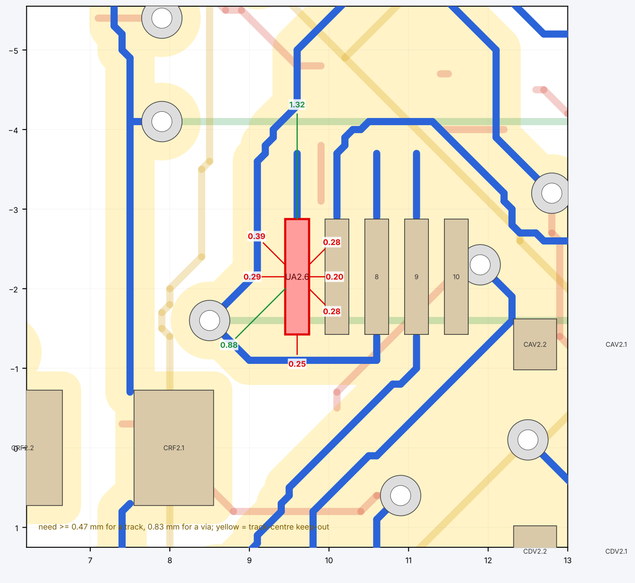
3 AB1A routed layout (best configuration A1): where it is open

board frame, y down; tracks coloured by layer; planes In1 = GND and In4 = +5VA are not drawn; red dashed = unrouted pad pairs



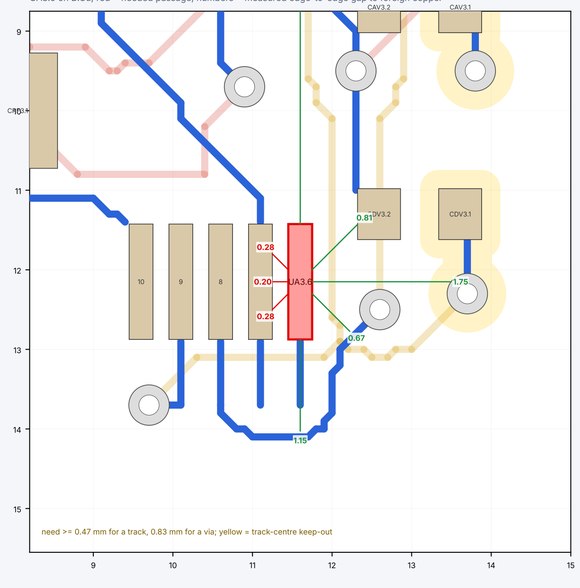
4 Open pad UA2.6 (CS)

UA2.6 on B.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



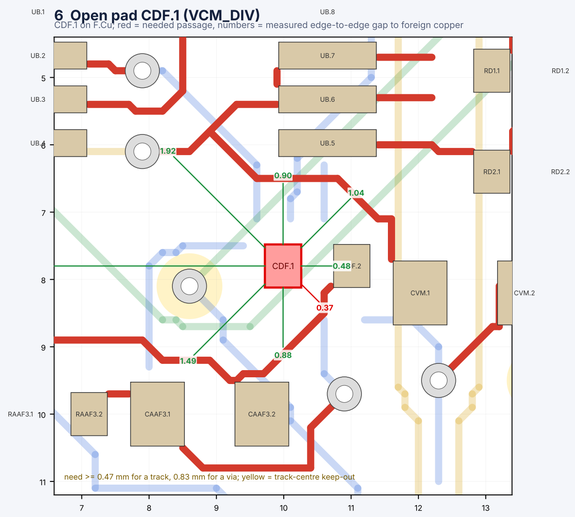
5 Open pad UA3.6 (CS)

UA3.6 on B.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



6 Open pad CDF.1 (VCM_DIV)

CDF.1 on F.Cu; red = needed passage, numbers = measured edge-to-edge gap to foreign copper



7 Why these pads cannot be reached, and every other open item

item	state / measured number	reason / next step
AB1A routing	OPEN: 3 nets (CS UA2.6; CS UA3.6; VCM_DIV CDF.1)	end pins of ADS8860 pin rows and 0402 pads in 0.5 to 1.0 mm pockets
ADS8860 pin row	pad 0.30 × 1.45 mm, pitch 0.50 mm: pad gap 0.20	a track needs 0.47 mm, a via 0.83 mm: nothing passes between pins
Fab floor (AB1 override)	clearance 0.16, track 0.15, via 0.508 / drill 0.254	TTM T-HDI Standard, AdvancedPCB A-TOL; U-FAB-DRC: some facilities list 0.216 ring
Router	greedy single pass A*, 0.1 mm grid, no rip-up	earlier nets wall in later pads; the open set moves, it does not vanish (A1 ↔ A2)
Shared corridor	3 ADCs at y pitch 10.0 mm, 2 pin rows each + decoupling + SPI + CS	about 4.2 mm between adjacent ADC rows shared by two 5-pin rows
J2 pad block	2 × 6, 1.27 mm pitch, pads 1.10 mm (ring 0.225, gap 0.17)	even column reachable only from outside; moving it is option (a): J2 to x = 17.5 mm with CB1D JA/JB
AB1B	routed, KiCad DRC 0 errors / 0 unconnected, 20 dangling-stub warnings	TODO: independent DRC zone exemption, Gerbers, publish, sign-off row
Datasheet pin maps	ADS8860, SW1G66, REF5041 marked VERIFY in afe_design.py	not checked against manufacturer datasheets
Mechanics	CORE cylinder + 14 other STLs non-manifold; U8_SEPTUM_extended oversize	vault +8 mm (option 2) and rim disc not implemented; no eddy / field model for an added board
Unverified	U-DS-TXDRIVE, OKI, PACK, BAV199, HVTERM, U-FAB-DRC, U-CAM-VIEW, U-BENCH	datasheets / fabricator DRC / CAM viewer / bench are not done